

The Stimulative Effect of State Tax Cuts Across the Income Distribution*

John Juneau[†]

Byron Lutz[‡]

October 31, 2025

Abstract

The ability of state governments to stimulate their economies through tax policy changes has long been debated. At the same time, a distinct debate considers the extent to which the distributional targeting of tax changes influences their effect on economic activity. This paper examines the intersection of these debates; i.e. we ask to what extent the distributional targeting of a state government income tax change mediates the effect on a state's economic activity. We do so using thirty years of detailed income tax return micro data and multiple panel data research designs—including an event-study local projection approach. Our results suggest that state-level employment responds positively to tax cuts targeted at the broad lower portion of the income distribution, but provide no evidence that economic activity responds to tax shocks targeted at the highest income households.

*We thank Katherine Richards for superb research assistance and Sutirtha Bagchi, Marcelo Bergolo, Timothy Godspeed, Alexander Klemm, and Nick Turner for useful discussions and suggestions.

[†]University of California-San Diego

[‡]Federal Reserve Board of Governors. All opinions expressed in the paper are solely those of the authors and not necessarily those of the Federal Reserve Board.

1 Introduction

There has long been a debate over the extent to which tax policy is an effective lever for state governments to promote economic activity; or, to frame the issue differently, if high levels of state taxation restrain economic growth. Much of this debate focuses on supply side responses to state taxation such as labor supply, capital investment, and cross-state migration decisions (e.g. [Agrawal and Tester, 2024](#); [Cassidy et al., 2024](#); [Giroud and Rauh, 2019](#); [Rauh and Shyu, 2024](#)). A related, but distinct, aspect of this debate focuses on the extent to which state governments can use tax policy to buffer the business cycle, in particular the extent to which tax policy can bolster shortfalls in consumer and business demand during economic downturns (e.g. [Carlino and Inman, 2013](#); [Clemens and Miran, 2012](#)).

A typically distinct debate, mostly focused on federal taxation, concerns how the distributional aspect of tax changes—i.e. where in the income distribution the incidence of a tax change falls—influences the extent to which the change affects economic activity (e.g. [Mertens and Montiel Olea, 2018](#); [Zidar, 2019](#)). For instance, low and middle-income households may have higher marginal propensities to consume than higher-income households; moreover, they may also face high marginal tax rates due to means-tested social insurance programs. If so, targeting tax cuts to the middle and lower end of the income distribution may spur economic activity more than a similar magnitude reduction aimed at the upper end of the income distribution. On the other hand, the marginal tax rates faced by high income households under progressive taxation may mean that targeting cuts to the high end of the distribution will be particularly effective at spurring investment, hiring, and labor supply. If so, high income tax cuts may “trickle down” and boost the broader economy (e.g. [Kindsgrab, 2022](#)). Finally, in the context of state income taxes—the focus of this paper—the relative effect of the tax cut is further mediated by possible differences in the geographic scope of behavioral responses by income. For instance, the propensity across the income distribution to purchase locally-produced, non-tradeables versus tradeables produced out of state will influence the economic response within a state.

This paper explores how the distributional targeting of a state tax change mediates the effect of the tax shock on a state’s economic activity; thus, it considers the intersection of the two debates above. An important conceptual distinction between this topic and related work focused on federal tax changes is the presence of balanced budget rules at the state level ([Poterba, 1994](#)). As a result of these budget rules, state tax changes are typically not deficit financed for any length of time, potentially altering their economic effects relative to federal tax changes which are often partially or fully deficit financed.¹ Moreover, the scope for geographic adjustment in response to state taxes is much greater than for federal taxes. Thus, results pertaining to distribution targeting of taxes at the federal level (e.g. [Zidar, 2019](#)) need not apply at the state level.

Our topic is of significant policy relevance given that state tax systems have very different relative distributional targeting – i.e. the progressivity differs greatly from state to state (e.g. [Cooper et al., 2015](#); [Fleck et al., 2021](#)). Moreover, states have been frequently shifting the distributional targeting of their income taxes in recent years. For example, since 2021 five states have transitioned to a flat tax, shifting the relative tax burden away from high earners, and a number of additional states are considering this change ([Walczak and Loughhead, 2024](#)). In contrast, other policy changes have increased state tax burdens on high incomes. In 2012, California significantly increased tax rates for high income households. 2017’s federal Tax Cut and Jobs Act effectively increased state income taxes for households with high income by capping the deductibility of state and local taxes on federal returns ([Gleckman, 2017](#)). The results of this paper help inform how these distributional changes in state tax policy influence the economic performance of a state.

Turning to our methodology, using the technique used to calculate federal tax shocks in [Zidar \(2019\)](#), we quantify state income tax shocks at different points in the income distribution using individual income tax data and the NBER’s TAXSIM tax calculator. Specifically,

¹E.g. if a state tax cut requires spending reductions due to balanced budget rules, the negative effect of these spending cuts on economic activity may partially or completely offset the positive economic effects of the tax reductions.

TAXSIM is used to calculate two tax liabilities for each year t tax unit: the year t liability based on the year t tax code and the year $t+1$ liability based on the year $t+1$ tax code. The difference in these liabilities is the tax unit’s “tax shock” for year $t+1$. Using year t data to estimate the year $t+1$ tax liability ensures that the shock measure solely captures changes in the tax code, not behavioral responses to the tax change. Finally, the tax-unit-year shocks are then aggregated into state-year-income-group tax shock measures. We use two income groupings: the top 10 percent of the income distribution and the bottom 90 percent of the distribution.

Our state tax shock data spans 30 years and provides rich empirical variation in terms of income tax changes. Moreover, the top 10 and bottom 90 income tax shock measures are not highly correlated. As a result, the data provide sufficient variation to identify separate effects for the two types of shocks.

We estimate the dynamic response of economic activity to these shocks using two panel data research designs. The first, and featured, approach is a local projection event study design. The second approach is a more sparse panel fixed-effect design utilizing long differences. We focus on employment as an outcome variable as it is well measured and consistently defined over a long period of time at the state level.

In terms of results, we find evidence that state income tax cuts among the bottom 90 percent of the income distribution boost economic activity notably. In contrast, our estimates for the effect of top 10 percent tax shocks are small in magnitude and imprecisely estimated, providing no evidence that cuts for high-earning households boost economic activity. Due to the presence of balanced-budget requirements, our estimates are properly viewed as net effects which capture the effect of the change in government spending, transfers, or other taxes that may accompany the initial income tax cuts. We also present estimates with extensive controls for changes in government spending and other taxes which attempt to isolate the effect of the income tax cuts. These estimates are essentially identical to our primary estimates.

An important concern with these estimates is the extent to which they reflect the causal effect of tax-cuts, as opposed to spurious correlation with unobserved factors. For instance, if state policy makers cut taxes during economic booms—say in order to return a surge in tax receipts to taxpayers—then our estimates may be biased. We address such concerns in multiple ways. First, we control for the state of the business cycle through geographic-by-year fixed effects (e.g. region-by-year) and other control variables. Notably, we directly control for state-specific budget shocks which might induce tax changes; this approach is similar in spirit to the narrative identification method of [Romer and Romer \(2010\)](#), but is more data driven. Moreover, our tax shocks are constructed to reflect only changes in tax policy parameters and hence will not mechanically be correlated with economic activity (i.e. a boost in economic activity will mechanically push up tax collections, but this rise in collections will not be reflected in our tax shock measure). Second, using local projection methods, we directly control for the recent, lagged evolution of the economic outcome variable being studied. Third, and perhaps most convincingly, we estimate event-study specifications which document that the effect of tax changes on middle and low income households materializes only *after* the tax change has been made. These specifications suggest that differential trends in economic activity across the states, correlated with tax adjustments, do not explain our results. Thus, they reduce, but do not eliminate, the scope for our results spuriously reflecting omitted variable bias. Finally, we rely on the comprehensive analysis of state tax changes in [Robinson and Tazhitdinova \(2023\)](#) which documents that such tax changes are largely idiosyncratic in nature and fail to correlate even moderately with observable factors such as economic conditions.

In future iterations of this project, which is work in progress, we have numerous refinements we intend to explore. These include using political-economy instrumental variables to further bolster a causal interpretation of the estimates. In addition, our event-study estimates currently require the assumption of treatment-effect homogeneity in order to avoid the recently documented issues with treatment effect estimates produced using two-way fixed

effect models (e.g. [Dube et al., 2023](#)). While we view the assumption of treatment effect heterogeneity as not unreasonable in our setting, going forward we will explore relaxing the assumption.

2 Relevant Literature and Contribution

Our findings contribute to an extremely large literature estimating the response of economic activity to changes in tax policy. This literature, which straddles the fields of macroeconomics and public finance, is much too large to summarize here. We therefore very selectively review a small portion of the existing work.

Starting with the literature relatively more grounded in macroeconomics, our paper is closely related to research estimating fiscal multipliers for taxes, transfers, and government spending. Early work on this topic includes [Andersen and Jordan \(1968\)](#), [Barro \(1981\)](#), [Blanchard and Perotti \(2002\)](#), and [Perotti \(1999\)](#).² Seminal work by [Romer and Romer \(2010\)](#) utilizes an innovative narrative method to identify plausibly exogenous changes in US federal tax policy. More recent fiscal multiplier contributions include [Ramey and Zubairy \(2018\)](#), [Chodorow-Reich \(2019\)](#), [Serrato and Wingender \(2016\)](#), [Dupor and Guerrero \(2017\)](#), [Nakamura and Steinsson \(2014\)](#), and [Boehm \(2020\)](#). A recent review of the fiscal multiplier literature is available from [Ramey \(2019\)](#).

Particularly relevant for this study, [Mertens and Montiel Olea \(2018\)](#) calculate effective marginal tax rate time series for different income groups and exploit variation across U.S. tax reforms to identify the effects of tax rate changes. They find that tax cuts for the top 1% of the income distribution raise real GDP and boost incomes outside the top 1%. Aggregate economic activity is also boosted by tax cuts targeted at the bottom 99%, although this effect takes several years to materialize.

²Many contributors have built on those early findings including [Barro and Redlick \(2011\)](#), [Ramey \(2011\)](#), [Favero and Giavazzi \(2012\)](#), [Mertens and Ravn \(2010\)](#), [Mountford and Uhlig \(2009\)](#), [Afonso and Sousa \(2012\)](#), [Kraay \(2012\)](#), [Christiano et al. \(2011\)](#), [Ilzetzki et al. \(2013\)](#), [Auerbach and Gorodnichenko \(2012\)](#), [Owyang et al. \(2013\)](#), [Woodford \(2011\)](#), [Corsetti et al. \(2012\)](#), and [Mertens and Ravn \(2010\)](#).

Turning to the literature relatively more grounded in the public finance tradition, our research is most closely related to the recent work by [Zidar \(2019\)](#), which studies the effects of federal tax shocks on economic activity conditional on whether shocks are targeted at the top or remainder of the income distribution. Conceptually, there are at least two important distinguishing features of our work relative to [Zidar \(2019\)](#). The first lies in the balanced-budget requirements of state governments: For a given tax cut, the potentially stimulative response to economic activity may be partially or completely mitigated by a corresponding decrease in government spending over time. Additionally, the boost to economic activity from an income tax cut may also generate additional tax revenues, thereby partially mitigating the need to reduce spending. Second, the cost for households and businesses of responding to a tax shock by relocating geographically is substantially lower for a state tax change than for a federal change; indeed, several very recent papers have highlighted that middle and particularly high income individuals may adjust geographically in response to state tax changes (e.g. [Agrawal and Tester, 2024](#); [Cassidy et al., 2024](#); [Giroud and Rauh, 2019](#); [Rauh and Shyu, 2024](#)). Such spatial adjustment may magnify the effect of a state tax shock relative to a federal tax change.

Another recent, relevant study is [Giroud and Rauh \(2019\)](#), which utilizes panel data methods and firm-level establishment data to examine the effects of state business taxation on business activity. They find that increases in marginal tax rates decrease employment and the number of establishments in a state. Key differences between this work and our paper include their focus on marginal tax rates, business taxation, and multi-state firms as compared to our focus on total tax burdens, personal income taxes (although these do include pass-through taxation of small businesses), and all employment within a state. Also relevant is recent research by [Carlino and Inman \(2013\)](#); [Clemens and Miran \(2012\)](#); [Suárez Serrato and Zidar \(2016\)](#) who document that states can use fiscal policy to stimulate economic activity. Finally, [Kindsgrab \(2022\)](#) evaluates distributional implications of changes in top marginal federal tax rates and finds no evidence that tax changes for high earners “trickle

down” to other workers.

3 Data

3.1 State Income Tax Shocks

To measure state income tax shocks, we utilize the IRS’ Statistics of Income Public-Use Files (PUF) for 1979-2009 alongside NBER’s TAXSIM calculator. We compute the year $t + 1$ shock for individual tax filer i in state s as the change in liability for year $t + 1$ relative to liability in year t holding year t income fixed. This calculation is given in equation 1 below where $L_{ist}^\tau = T(R_{s\tau}, I_{it})$ represents the mapping of the tax schedule $R_{s\tau}$ for year τ in state s and income I_{it} for year t to tax liabilities.

$$Shock_{ist+1} = L_{ist}^{t+1} - L_{ist}^t \quad (1)$$

With this formulation, a positive individual shock (i.e., $Shock_{ist+1} > 0$) reflects an increase in liability (i.e., “tax hike”) while a negative shock reflects a decrease in liability (i.e., “tax cut”).

The PUF data does not provide any information on individual consumption or savings; hence, we cannot use these shocks to estimate individual-level responses to the shock across the income distribution. Instead, we aggregate the individual tax shocks to two income-based categories for every year and state: the bottom 90% (B90) and top 10% (T10) of the national income distribution. For ease of interpretation, we also scale the aggregate tax shocks as a percent of state GDP. Hence, our final aggregate tax shock measures are given by:

$$Shock_{s't}^{X'} = \frac{100}{GDP_{st}} \sum_i \mathbb{1}_i[s = s'] \mathbb{1}_i[X = X'] Shock_{ist} \quad (2)$$

for $X \in \{B90, T10\}$.

We calculate aggregate tax shocks for every US state between 1980 and 2009 and present

summary statistics of these measures in Table 1 and Table 2. As shown in Table 1, the average state income tax shock incurred by the bottom 90% of tax-filers was a tax *cut* equal to 0.01% of state GDP. Meanwhile, the average T10 shock was a 0.001% tax *hike*. Both measures exhibit considerable variation with the B90 shocks ranging from -0.65% of state GDP to 0.54% and the T10 shocks ranging from -0.75% to 0.63%. To place these shocks into context, in the 1995 fiscal year total personal income tax collections across all state governments equalled about 1.5% of U.S. GDP. Table 2 displays selected percentiles of the tax shocks.

Table 1: Tax Shocks, Summary Statistics

	mean	sd	min	max
Bottom 90%	-.0112901	.0639122	-.6519895	.5379407
Top 10%	.0011699	.0572673	-.750378	.6326351
Observations	1530			

Table 2: Tax Shocks, Selected Quantiles

	p1	p5	p25	p50	p75	p95	p99
Bottom 90%	-.2142843	-.0831217	-.0185298	-.0012051	0	.0452868	.1777679
Top 10%	-.1277249	-.0571094	-.0043585	-7.40e-07	0	.0763605	.1965095
Observations	1530						

In Figure 1 below, we plot the yearly censored distributions of both B90 and T10 shocks from 1980 to 2009. We observe visually that both measures exhibit considerable variation in any given year, especially in the first decade of the sample.

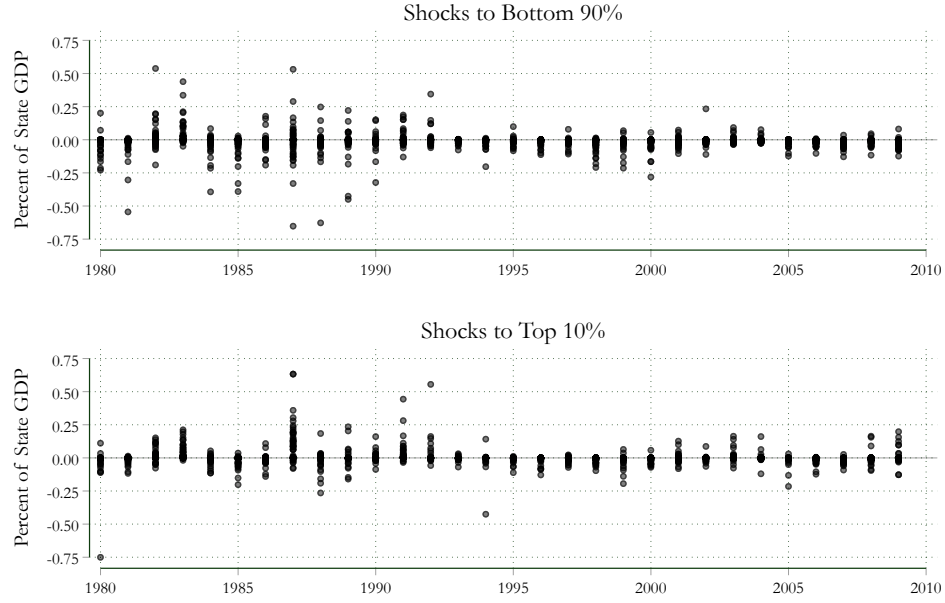


Figure 1: State Income Tax Shocks, 1980-2009

In order to identify separate effects of these aggregate tax shocks, the shocks cannot be perfectly correlated with each other. Conceptually, it is unlikely that this would be the case, but we can test for this statistically. Visually, we see in Figure 2 below confirmation that the shocks are far from perfectly correlated. Overall, a simple regression of the T10 shocks on the B90 shocks yields an adjusted R^2 of just 0.31. The unconditional correlation between the measures is 0.56.

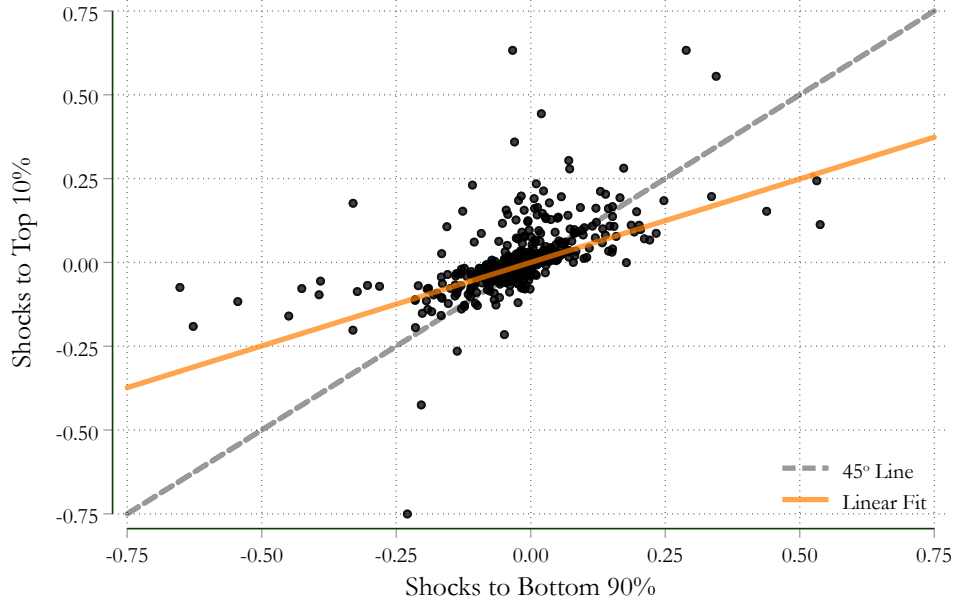


Figure 2: Correlation Between B90 and T10 Shocks

3.2 Economic Variables

Our primary outcome variable is total state-level employment as measured by the Bureau of Labor Statistics (BLS). In our regression analysis to follow, we evaluate the *growth* response of employment to the aggregate tax shocks described above. Accordingly, we provide summary statistics for employment growth in Table 3 below. In this period, employment grew by 1.2% annually on average. Of course, we also observe substantial heterogeneity, with employment growth ranging from -6.5% to 10.9%.

We also present very limited results using state-level GDP. We do not focus on GDP for two reasons. First, its measurement at the state-level entails significant amounts of imputation and many assumptions; as a result, it is not as well measured as state-level employment. Second, and more significantly, the measurement of state-level GDP underwent a major methodological redesign in 1997—a point in time which nearly cleaves our sample

into halves.³ The redesign was significant enough that it may have an important influence on both the level and growth rates of GDP; the Bureau of Economic Analysis—the statistical agency which produces the data—“strongly caution”s against appending the data together (BEA, 2020). Nevertheless, we present a few estimates based on state GDP.

Table 3: Economic Variables, Summary Statistics

	mean	sd	min	max
Output	2.259874	3.977487	-28.02676	29.38097
Employment	1.215534	1.97581	-6.546711	10.90028
Observations	1500			

In Figure 3 below, we plot the distributions of state output and employment growth from 1980 to 2009. We see clearly that economic activity varies considerably across states in any given year.



Figure 3: State Output and Employment Growth, 1980-2009

³The redesign shifted the measurement from being based on the SIC industry codes to the NAICS codes; it also conceptually shifted from being consistent with national GDI (gross domestic income) to national GDP. In addition, there are a number of differences in source data and estimation methodologies pre and post 1997.

4 Identification and Research Designs

4.1 Econometric Identification

The aim of this study is to evaluate how economic activity—especially employment—responds to changes in state income tax policy. In particular, we are interested in estimating whether those effects depend on the targeting of the policy shock: either the top or remainder of the income distribution. In principle, we can imagine estimating a regression of the following form to elucidate these effects.

$$y_{s,t} - y_{s,t-1} = \beta^{T10} RevShock_{s,t}^{T10} + \beta^{B90} RevShock_{s,t}^{B90} + \epsilon_{s,t} \quad (3)$$

where $RevShock_{s,t}^X$ is the tax revenue shock in state s in year t due to a change in collections from income group X . In order for our desired estimates to be identified by equation 3, it must be the case that $Cov[RevShock^X, \epsilon | RevShock^{X'}] = 0$; that is, our shock measures, $RevShock^X$, must be uncorrelated with the error term, ϵ , conditional on covariates. However, there are several reasons that this is unlikely to be true.

Notably, an analysis measuring tax shocks with changes in tax *revenues* is likely to be biased by simultaneity. That is, changes in revenues will themselves reflect changes in economic activity that are not the result of actual policy changes. For example, we can imagine a booming economy that features rising incomes and therefore rising tax liabilities. If we were to measure shocks as the change in revenues, then we would mistakenly attribute some economic growth to an increase in taxes when in fact the rise in taxes is the result of rising economic activity. Fortunately, our TAXSIM-estimated tax liability shock for year $t + 1$ is constructed holding income fixed at time t , with only the parameters of the state tax code varying from t to $t + 1$. This allows us to avoid this form of simultaneity bias. That said, other forms of omitted variable bias may persist.

To see this, suppose that our Taxsim-based shock measures, $Shock_{s,t+1}^X$, exclusively re-

flect changes in tax policy implemented in year $t + 1$ to stimulate economic activity in the wake of an economic recessions experienced in year t . In this setting our estimates may be biased downward due to correlation between the tax shock measure and decline in economic activity caused by factors other than tax policy. To the extent that such omitted factors are time-varying but constant across states, we could, for example, include year fixed effects to mitigate this concern. However, if the omitted factors vary across time and state, then our estimates may be biased.

We adopt a number of approaches to address omitted variable bias. First, we control for various forms of year fixed effects allowed to vary by state characteristics—e.g. year-by-cyclical-sensitivity quintile fixed effects and year-by-census-region. Second, we directly condition on lags of the outcome variable—e.g. three lags of employment growth. This in principle addresses the concern that state policy makers adjust tax policy in response to state-year specific changes in the economic outcomes being studied.⁴ Finally, and perhaps most persuasively, we estimate event-study specifications. The estimation of pre-tax shock treatment effect vectors allows us to assess if tax shocks are correlated with state-specific trends in economic activity that precede the tax shock. Such trends would likely bias our estimates.

The focus of our study is state income tax shocks, which occur in a context of balanced-budget requirements. This complicates the interpretation of our results. Suppose that a government implements a tax cut to boost economic activity by giving households more disposable income to consume. For simplicity, suppose that the state generates all revenues from its income tax. Hence, the immediate impact of the policy change would, all else equal, create a budget deficit as revenue falls but expenditures remain constant. If balanced-budget requirements are sufficiently strict, then the state must accompany its tax cut with a corresponding cut in expenditures. From the perspective of equation 3, this generates what could be viewed as omitted variable bias—i.e. the failure to control for the effect of *spending*

⁴State policy makers set tax policy for year $t + 1$ in year t . Thus, conditioning on lagged employment controls for the observable trajectory of economic conditions at the time tax policy for year $t + 1$ is determined.

shocks. Alternatively, the coefficients on the revenue shock measures in equation 3 could be viewed more broadly as net effects which capture both the tax cuts and the spending cuts necessitated by the tax cuts.

The reality is, of course, much more complicated. States’ balanced-budget requirements allow for substantially more flexibility than the scenario described above.⁵ States also generally do have other sources of revenue including sources that may yield higher revenues in the face of an income tax cut. Moreover, the revenue from income taxes themselves may increase over time if the tax cut boosts economic activity and therefore leads to larger tax base through higher employment or wages. Consequently, our primary estimates should be interpreted as the “all-in” effect of a tax shock that includes the effects of any expenditure changes necessitated or allowed by the tax change. That said, we also estimate specifications with extensive measures of state government spending which attempt to control for any systematic changes in spending caused by state tax changes and thereby isolated the effect of the tax change.

A final source of possible omitted variable bias arises from changes in aspects of state tax policy other than the personal income tax. For instance, if income tax rates are cut, then corporate tax rates might be increased to cover the reduced revenue from personal income tax collections. More generally, [Robinson and Tazhitdinova \(2023\)](#) and [Baker et al. \(2024\)](#) document that state tax changes often occur in tandem, with multiple tax instruments being changed at the same time. Accordingly, we present specifications which control for changes in the state sales tax rate, state corporate income tax rate, and, per [Giroud and Rauh \(2019\)](#), state and local government property tax collections as a share of total state and local government revenue.

Finally, the credibility of our econometric identification is bolstered by the comprehensive analysis of state tax changes in [Robinson and Tazhitdinova \(2023\)](#) which documents that

⁵States have a number of methods with which to smooth expenditures over the short run—e.g. drawing down rainy day funds. Nevertheless, it is well documented that balanced budget rules bind (e.g [Poterba, 1995, 1994](#); [Clemens and Miran, 2012](#)) and significantly constrain state fiscal policy. But they need not strictly and perfectly bind on a year-to-year basis.

such tax changes are largely idiosyncratic in nature; they fail to correlate even moderately with observable factors such as economic conditions.

4.2 Research Design

We utilize two specifications to evaluate the relationship between state income tax shocks and economic activity. These specifications are similar to those employed in [Zidar \(2019\)](#).

We first utilize a very general direct projection model that regresses the difference in log level of employment in a year $t + h$ and $t - 1$ for horizon $h \in \{-3, -2, \dots, 3, 4\}$ on year t tax shocks.

$$y_{s,t+h} - y_{s,t-1} = \alpha_h^{B90} Shock_{s,t}^{B90} + \alpha_h^{T10} Shock_{s,t}^{T10} + \mathbf{X}'_{st} \boldsymbol{\Lambda}^h + \eta_{s,h} + \gamma_{cs,t,h} + \epsilon_{s,t,h} \quad (4)$$

where $\eta_{s,h}$ are state fixed effects allowed to vary by h and $\gamma_{cs,t,h}$ are cyclical-sensitivity quintile(cs)-by-year(t) fixed effects allowed to vary by h . The cyclical-sensitivity quintiles are time invariant and are estimated using the procedure in [Blanchard et al. \(1992\)](#). Intuitively, the cyclical-sensitivity quintiles group states by the responsiveness of their economic activity to changes in national economic activity.

Included in the $X_{s,t}$ vector are a full set of leads and lags of $Shock_{s,t}^{B90}$ and $Shock_{s,t}^{T10}$. The leads and lags are needed to control for the dynamic effects of previous shocks (occurring at $t < 0$) and future shocks (occurring at $t > 0$) under the assumption of treatment effect heterogeneity ([Dube et al., 2023](#); [Sun and Abraham, 2020](#)).⁶

The parameters of interest in this regression are $\hat{\alpha}_h^{B90}$ and $\hat{\alpha}_h^{T10}$. This event study design permits us to examine “pre-trends” (i.e., estimates for $h < 1$), which will help to illuminate the credibility of our econometric identification.

⁶We assume that the full dynamic effect of a shock manifests within five years, inclusive of the year of the shock; thus, four lags must be controlled for when estimating the post-treatment vector. Leads must also be controlled for when estimating the post-treatment coefficients to account for the effect of shocks occurring at $t > 0$. As an example, at horizon $h=1$ we control for four lags of the tax shock measures and one lead of the tax shock measures; at horizon $h=2$ we control for four lags of the tax shock measures and two leads of the tax shock measures.

Second, we utilize a much more parsimonious specification that regresses the four-year growth in output or employment between year t and $t - 4$ on the cumulative shocks over those years.

$$\frac{Y_{s,t} - Y_{s,t-4}}{Y_{s,t-4}} = b^{B90} \left(\sum_{m=0}^3 Shock_{s,t-m}^{B90} \right) + b^{T10} \left(\sum_{m=0}^3 Shock_{s,t-m}^{T10} \right) + \mathbf{W}_{st}' \mathbf{\Lambda} + a_s + d_t + \epsilon_{s,t} \quad (5)$$

Across all specifications we cluster standard errors by state. We do not weight the specifications so that each state provides an independent and equally weighted "experiment." In most instances, we estimate our specifications on our full 1980-2009 sample.⁷

Both our research designs are forms of two-way fixed effect (TWFE) models. Given that our tax shock treatments are staggered and have dynamic effects, a TWFE model may fail to recover policy relevant treatment effects if the treatment effects are heterogeneous across states (e.g. [Callaway and Sant'Anna, 2020](#); [Sun and Abraham, 2020](#); [Goodman-Bacon, 2021](#); [de Chaisemartin and D'haultfœuille, 2022](#); [Roth et al., 2023](#); [Dube et al., 2023](#)). We take three steps with respect to our research designs in response to this concerns.

First, for our primary results we impose the assumption that treatment effects are homogeneous. All empirical research designs require the imposition of assumptions in order to recover interpretable parameters and we view the assumption of treatment effect heterogeneity as reasonable in our setting. Indeed, as emphasized in [Arkhangelsky and Imbens \(2023\)](#), "in many cases fixed effect models with constant treatment effects may be good working models that lead to robust estimates and inferences."⁸ Moreover, our research designs are complex—e.g. the direct projection approach involves simultaneously estimating two dynamic treatment effects with strictly continuous, non-absorbing treatments that take on non-positive values. This richness naturally requires imposing more stringent assumptions

⁷The PUF micro tax data lacks state identifiers beginning in 2010.

⁸In contrast, treatment effect heterogeneity is a first-order concern in a setting in which units treated later in the sample period are likely to systematically and for identifiable reasons have different magnitude treatment effects than those treated earlier in the sample period. See [Autor et al. \(2022\)](#) for an example where this condition appears to hold. Our setting is arguably one with much less obvious and clearly identifiable reasons for treatment effect heterogeneity.

than a less rich specification. Furthermore, important and prominent related work done in a time-series setting—e.g. [Mertens and Montiel Olea \(2018\)](#)—requires even more stringent assumptions than those employed here.

Second, in order to assess the likelihood of treatment effect heterogeneity, we explicitly test for it. For instance, the fiscal multiplier often discusses treatment effect heterogeneity as a function of the amount of slack in the economy and by positive versus negative fiscal shocks. Accordingly, we explicitly test for such heterogeneity. We find little evidence of treatment effect heterogeneity along a number of dimensions.

Third, in future work we will explicitly relax the treatment effect heterogeneity assumption using the "clean control" techniques developed in [Dube et al. \(2023\)](#). Doing so, however, will require a very significant reduction in sample size and it is therefore unclear how informative the results will be.

5 Results

We first present the estimated results from the direct projection specification, equation 4, in Figure 4 below. The solid lines denote the estimated coefficients (blue for B90 and red for T10) while the dashed lines denote standard errors around each estimate. By construction, the estimates $t = -1$ years since the tax shock is equal to zero. For $t \geq 0$ years since the shock, we see that employment responds negatively to B90 shocks—i.e. a tax cut targeted at the bottom 90 percent of the income distribution causes a boost in employment, while a tax increase aimed at this group reduces employment. A tax cut of 0.25% of state GDP targeted at the bottom 90% of the national income distribution translates into about 2 pp. higher employment growth by year $t = 2$; the effect is relatively constant thereafter. The estimated pre-trend for bottom 90% tax shocks is relatively flat and imprecisely estimated, with small point estimates. This finding is consistent with a causal interpretation of the post-treatment bottom 90% tax shock estimates (although it fails to prove causality).

In contrast to the treatment effect estimates for tax shocks targeted at the bottom 90% of the income distribution, the estimates for tax shocks targeted at the top 10 percent of the income distribution are uniformly very small and statistically indistinguishable from zero. Moreover, the estimates for the bottom 90% and top 10% can be statistically distinguished from one another for $t = [0, 1, 2]$ at the 0.021, 0.055, and 0.079 confidence levels, respectively. Thus, our results suggest that employment growth responds positively to B90 tax cuts, but is unresponsive to T10 cuts.

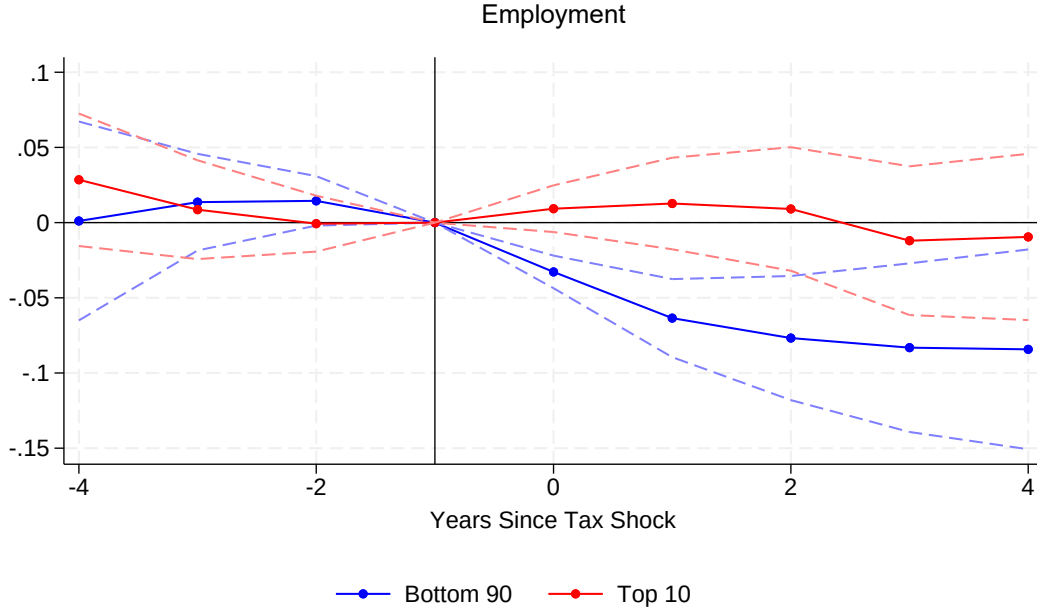


Figure 4: Direct Projection Results

Note: Estimated using equation 4. Dependent variable is the horizon h log difference of employment. Specification includes state fixed effects year-by-cyclical-sensitivity quintile fixed effects, all allowed to vary by horizon h ; it also includes controls for federal tax shock as defined in Zidar (2019).

Figure 5 displays the results of the estimation the direct projection specification with controls for the lagged dependent variable. Specifically, $y_{s,t-1} - y_{s,t-2}$, $y_{s,t-2} - y_{s,t-3}$, and $y_{s,t-3} - y_{s,t-4}$ are included as controls (with employment y in logs); these are the three most recent annual rates of employment growth observable to policy makers at time $t - 1$ when they are setting time t tax policy. Including these variables as right-hand side controls precludes estimation of the pre-treatment portion of the treatment effect vector. Results for

the post-treatment portion of the treatment effect vector are essentially unchanged from the results in figure 4 which omit the lagged dependent variables.

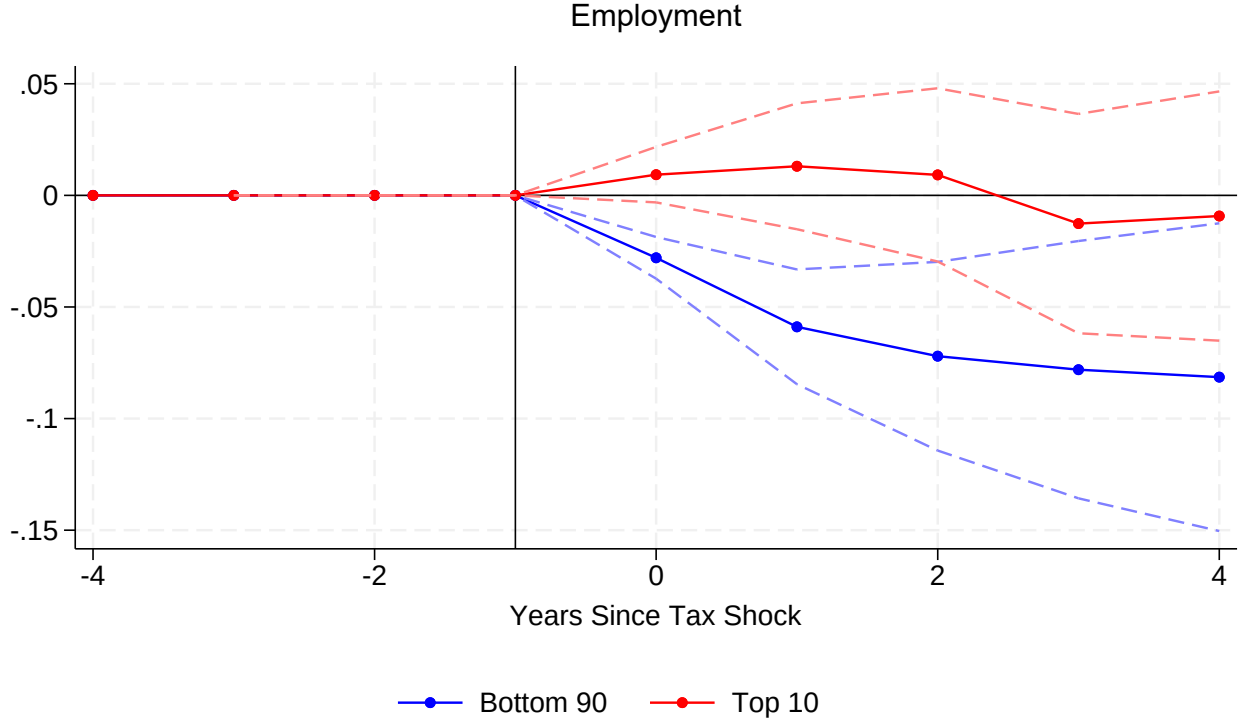


Figure 5: Direct Projection Results, Controls for Lagged Dependent Variables

Note: Estimated using equation 4. Dependent variable is the horizon h log difference of employment. Specification includes state fixed effects, year-by-cyclical-sensitivity quintile fixed effects, and three lags of the dependent variable, all allowed to vary by horizon h ; it also includes controls for federal tax shock as defined in Zidar (2019).

Figure 6 presents the results of additional alternate specifications. In order to ensure our results are not driven by a few extreme outliers in the state tax shock distributions, Panel A estimates a specification in which the tax shocks have been truncated such that they do not exceed 0.5 percent of GDP or fall below -0.5 percent of GDP. The results are little changed.

In Figure 4, the individual coefficients are not all identified by the same set of tax shocks. This arises due to the tax shock leads and lags included in the $X_{s,t}$ vector in equation 4. For instance, the horizon $h = 0$ coefficient α_0^{B90} obtains identification from a tax shock occurring in 2009—the last year we have tax shocks available. However, the horizon $h = 1$ coefficient α_1^{B90} does not obtain identification from the 2009 tax shock because of the tax shocks lead

being controlled (which would require a 2010 tax shock be observable). Accordingly, it is possible that the change in the treatment effect as a function of h spuriously arises from the differing set of tax shocks identifying the parameters—i.e. the estimates may suffer from composition effects. The specification presented in panel B therefore restricts the sample such that all treatment effect coefficients h are identified from the same set of shocks. The results are broadly unchanged from Figure 4.

Panel C includes controls for changes in the state sales tax rate and the top marginal state corporate income tax rate. Panel D includes controls for government expenditures; specifically, the log difference of real total expenditures of state and local government in state s at time t is controlled for. Panel E includes both the tax controls used in Panel C and the expenditure controls used in Panel D. Finally, Panel F includes the tax controls in Panel C and also includes state and local government property tax collections as a share of total state and local government revenue per the approach of [Giroud and Rauh \(2019\)](#). Furthermore, an expanded set of government real expenditures are controlled for, including state government current operational expenditures, state government capital expenditures, state government assistance and subsidy payments, local government current operational expenditures, and local government capital expenditures. In all cases, the results are broadly similar to the primary direct projection specification presented in Figure 4. Thus, the results which capture the net effect of the personal income tax shocks, inclusive of balanced-budget induced offsetting expenditure and non-personal income tax changes (Figure 4), are similar to the results which attempt to isolate only the effect of the personal income tax shocks (Panels C-F of Figure 6). One way to reconcile this finding is that the offsetting tax and expenditure changes may have limited effect on economic activity—e.g. the expenditure multiplier may be small.

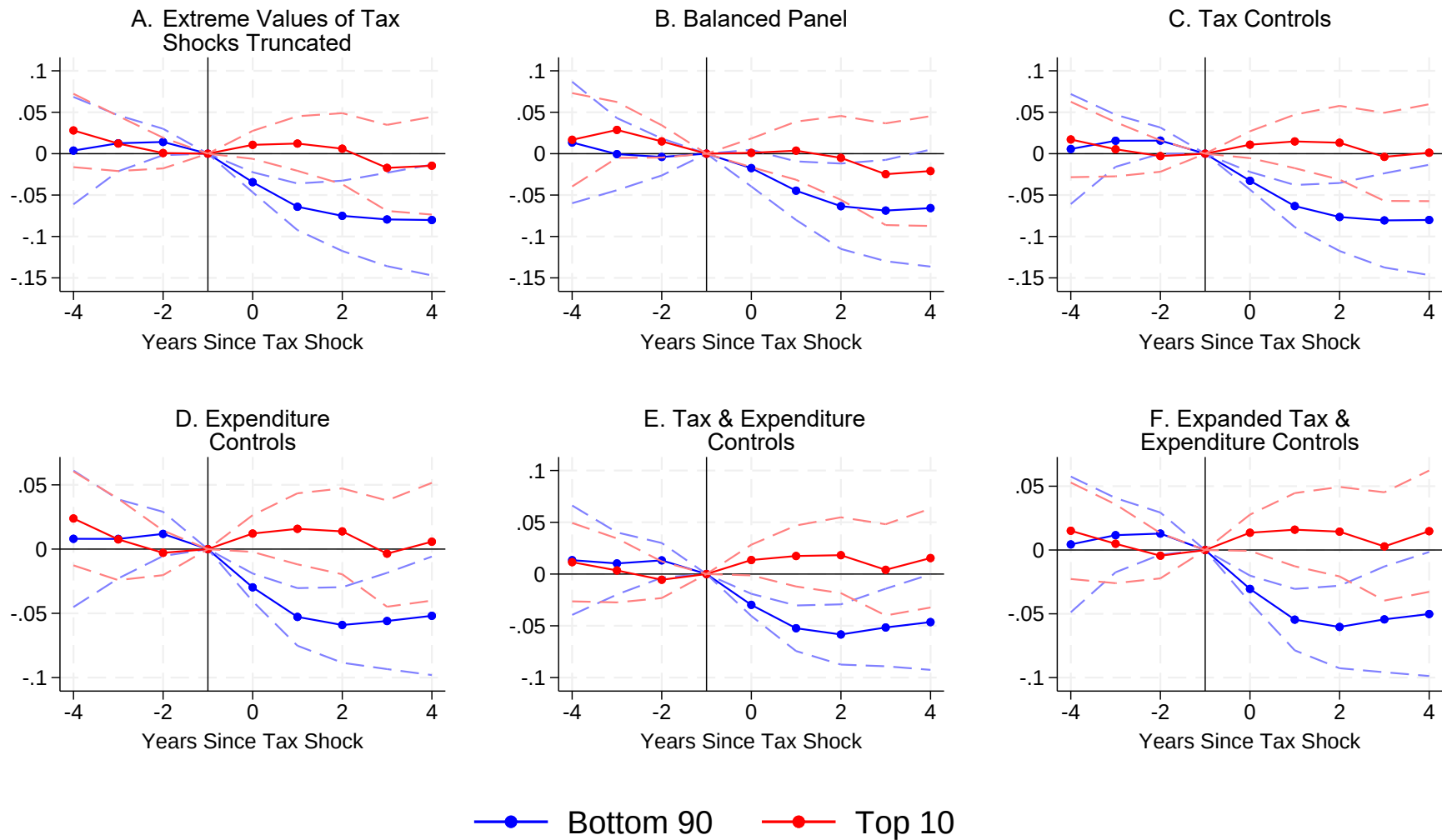


Figure 6: Direct Projection Results, Employment Alternative Specifications

Note: Estimated using equation 4. Dependent variable is the horizon h log difference of employment. Specifications include state fixed effects and region-by-year effects, all allowed to vary by horizon h ; it also includes controls for federal tax shock as defined in Zidar (2019).

We turn now to the estimated coefficients from the more parsimonious four-year growth specifications, equation 5. These are presented in Table 4. The columns use various controls. The first column uses controls which mimic those used in Figure 4. The second column replaces the cyclical sensitivity quintile-year fixed effects with region-year fixed effects. Some readers may prefer the regions-year effects because, unlike the cyclical sensitivity quintile-year fixed effects, they do not require estimation to form the groupings by which the year effects are allowed to vary; instead, they use a long-established geographical grouping. The third column again uses the cyclical sensitivity quintile-year fixed effects and adds a distinct year effect for oil producing states.⁹

The final column controls for lagged deficit shocks—i.e. it captures the extent to which a state’s fiscal position improves or erodes relative to what was expected at the time that the fiscal year’s budget was set. For instance, if a state has an unexpected economic downturn, its tax receipts will fall short of what the state forecasted at the start of the fiscal year; in this case the state will experience a positive deficit shock. The measure is calculated using data from the National Association of State Budget Officers and calculated using the methodology of [Poterba \(1994\)](#) and [Clemens and Miran \(2012\)](#). These shocks are state-specific and controlling for them sheds light on the possibility that the results presented so far spuriously reflect state-year specific evolution in economic activity correlated with tax changes—e.g. that states are adjusting tax policy to address budget difficulties. Stated differently, conditional on the control for lagged deficit shocks, the tax shock measures can arguably be viewed as “exogenous” as they are not associated with efforts to address budget difficulties. Thus, they can be seen as very roughly analagous to the exogenous tax shocks of [Romer and Romer \(2010\)](#), but constructed in a more data driven manner.

Across all the columns the results are broadly similar to those presented above based on the direct projection method. Thus, the results appear to be reasonably robust to a variety of additional controls.

⁹Oil producing states are defined based on 1985 per-capita production: AK, WY, ND, NM, TX, and OK.

Table 4: Four-Year Change Model

Employment				
Top 10% Shock	-0.0160 (0.0220)	0.00774 (0.0223)	-0.0108 (0.0226)	0.0176 (0.0257)
Bottom 90% Shock	-0.0634*** (0.0144)	-0.0552*** (0.0183)	-0.0608*** (0.0152)	-0.0708*** (0.0250)
Lagged Deficit Shock				-0.0364*** (0.0103)
Observations	1350	1350	1350	807
R^2	0.737	0.719	0.745	0.826
Federal Shock Controls?	Yes	Yes	Yes	Yes
State FE?	Yes	Yes	Yes	Yes
Cyclical Quintile-Year FE?	Yes	No	Yes	Yes
Region-Year FE?	No	Yes	No	No
High Oil-Year FE?	No	No	Yes	No

Estimated using equation 5. Standard errors clustered by state in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5 contains results which explore treatment effect heterogeneity. These results are interesting in their own right. They also shed light on the concern that our TWFE models may fail to recover policy relevant treatment effects due to treatment effect heterogeneity.

We consider treatment effect heterogeneity across several dimensions. The first two dimensions are whether the shock is positive versus negative and the amount of slack in the economy (as measured by the national unemployment rate). These are motivated by the focus on possible heterogeneity along these dimensions in the fiscal multiplier literature. Finally, we also consider heterogeneity by the size of the tax shock (addressed by including the shock times its absolute value—i.e. the sign-preserving square of the shock.)

We find no evidence of heterogeneity by positive versus negative shocks or by the size of the shock. While there is some evidence of heterogeneity with respect to slack in the economy—see the third column—the interaction terms have little effect on the main tax shock coefficients. Moreover, controlling for the deficit shocks renders the interaction terms very imprecise. Thus, we find little evidence in support of treatment effect heterogeneity.

Table 5: Four-Year Change Model

Employment					
Top 10% Shock	-0.016 (0.022)	-0.022 (0.035)	-0.002 (0.018)	0.015 (0.025)	-0.001 (0.027)
Bottom 90% Shock	-0.063*** (0.014)	-0.041** (0.018)	-0.069*** (0.016)	-0.064** (0.025)	-0.088*** (0.027)
Top 10% Shock * (Shock > 0)		0.013 (0.046)			
Bottom 90% Shock * (Shock > 0)		-0.049 (0.039)			
Top 10% Shock * Nat. Unemp. Rate			-0.061*** (0.016)	-0.014 (0.015)	
Bottom 90% Shock * Nat. Unemp. Rate			0.030** (0.014)	0.023 (0.019)	
Top 10% Shock * Top 10% Shock					-0.025 (0.039)
Bottom 90% Shock * Bottom 90% Shock					0.058 (0.048)
Observations	1350	1350	1350	807	1350
R^2	0.737	0.750	0.744	0.828	0.740
Federal Shock Controls?	Yes	Yes	Yes	Yes	Yes
State Fixed Effects?	Yes	Yes	Yes	Yes	Yes
Cyclical Quintile-Year Fixed Effects?	Yes	Yes	Yes	Yes	Yes
Lagged State-Specific Deficit Shock?	No	No	No	Yes	No

Estimated using equation 5. Standard errors clustered by state in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

For completeness, Table 6 presents results for the employment-to-population ratio * 100 (Panel A) and real GDP (Panel B). The dependent variable in Panel A is the four-year difference in the employment-to-population ratio (as opposed to the percent change in equation 5 and used in Panel B). The results again suggest that a tax cut aimed at the broad lower part of the income distribution spurs economic activity, while there is no such evidence for cuts targeted at the top of the income distribution. As discussed above, though, the GDP results are quite speculative given the methodological redesign for state-level GDP in the middle of the sample period.

Table 6: Four-Year Change Model

Panel A: Employment-to-Population * 100		
Top 10% Shock	-0.714 (0.939)	0.161 (0.738)
Bottom 90% Shock	-1.891** (0.794)	-1.381* (0.749)
Observations	1350	1350
R^2	0.679	0.660
Federal Shock Controls?	Yes	Yes
State FE?	Yes	Yes
Cyclical Quintile-Year FE?	Yes	No
Region-Year FE?	No	Yes
Panel B: Real GDP		
Top 10% Shock	-0.0332 (0.0444)	0.0461 (0.0487)
Bottom 90% Shock	-0.0959*** (0.0334)	-0.110** (0.0454)
Observations	1192	1192
R^2	0.559	0.422
Federal Shock Controls?	Yes	Yes
State FE?	Yes	Yes
Cyclical Quintile-Year FE?	Yes	No
Region-Year FE?	No	Yes

Estimated using equation 5 with the exception that in Panel A, the dependent variable is defined as $Y_{s,t} - Y_{s,t-4}$.

Standard errors clustered by state in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

6 Conclusion

How does economic activity respond to changes in state income taxes? Do those responses depend on the targeting of the policy change? In this study, we provide evidence that tax cuts among the bottom 90% of the income distribution generate moderate increases in economic activity—specifically employment—while shocks among the richest households generate no such effect. These results contribute to a large literature on fiscal multipliers and the effects of tax policy changes.

As we continue to develop this research, we hope to further refine our research design in a number of ways. Items we view as a high priority include, but are not limited to, using political-economy instrumental variables to further bolster a causal interpretation of the estimates. In addition, our event-study estimates currently require the assumption of treatment-effect homogeneity in order to avoid the recently documented issues with treatment effect estimates produced using two-way fixed effect models (e.g. Dube et al., 2023). Going forward, we will relax the assumption of treatment-effect homogeneity.

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